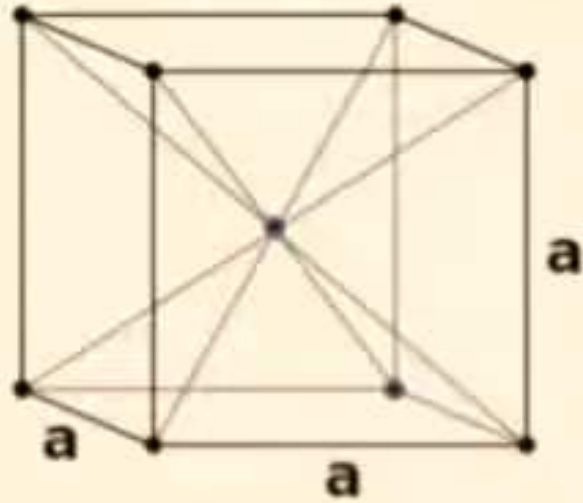




# Unit cell of Body Centered Cubic Lattice (BCC) (Notated cubic-I)

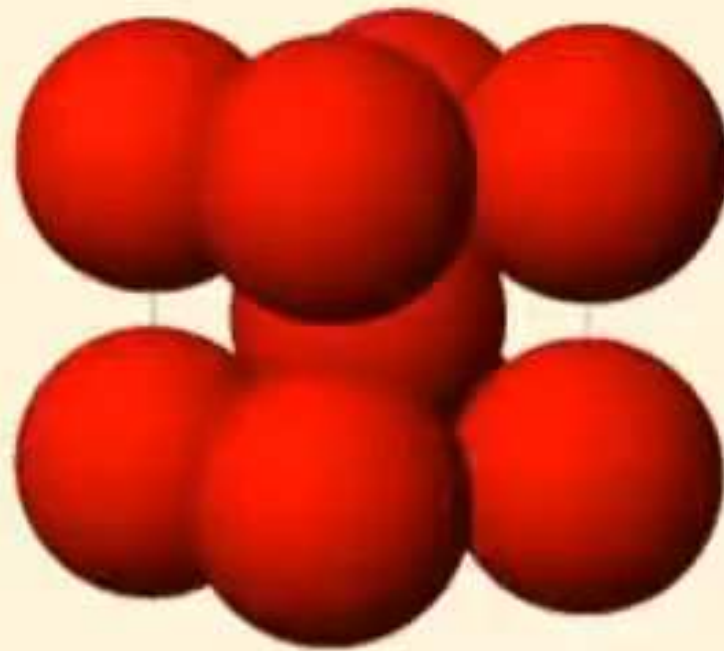


Conventional Unit Cell



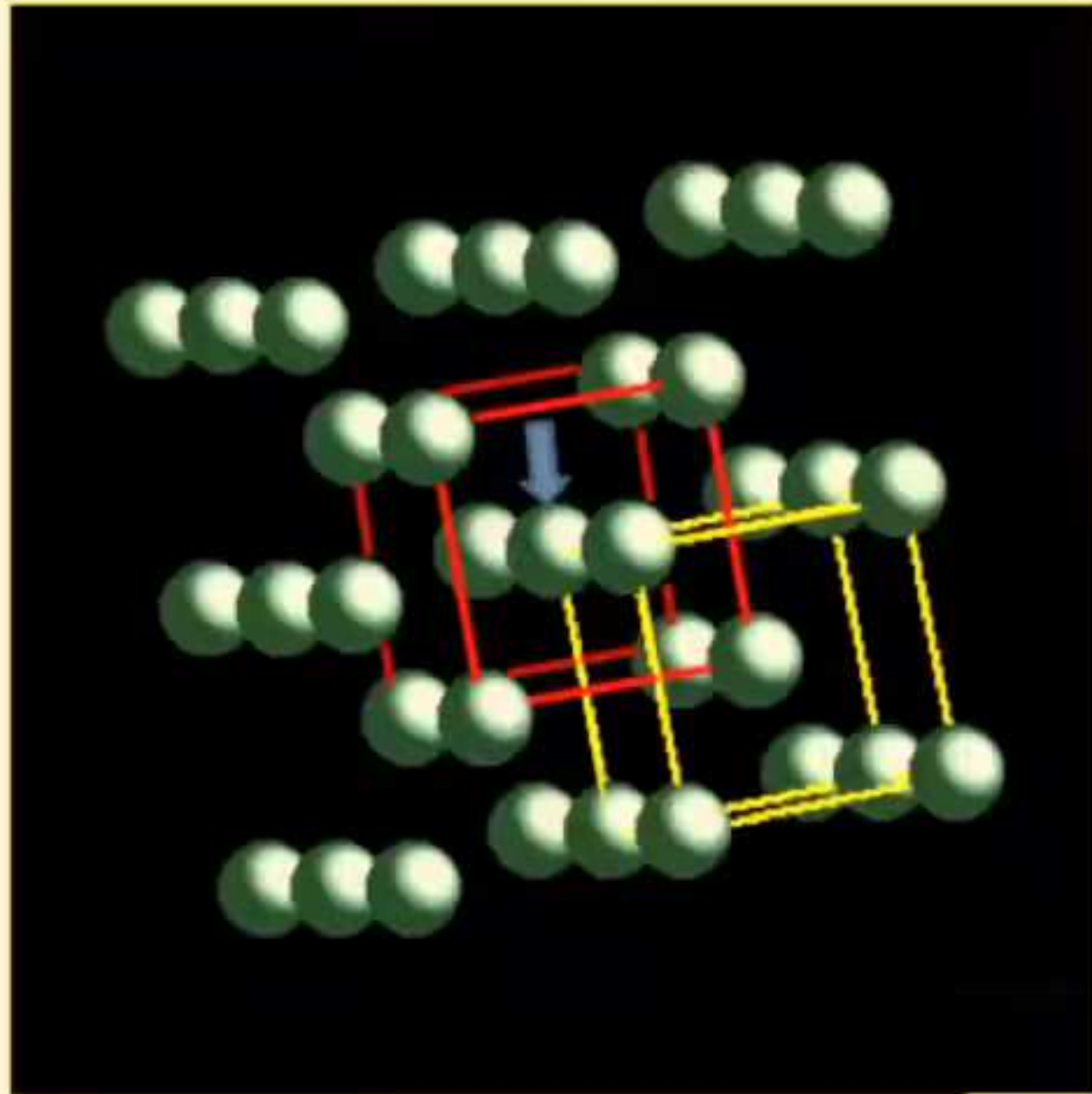
Plan View

(unlabeled points at height 0 and a)



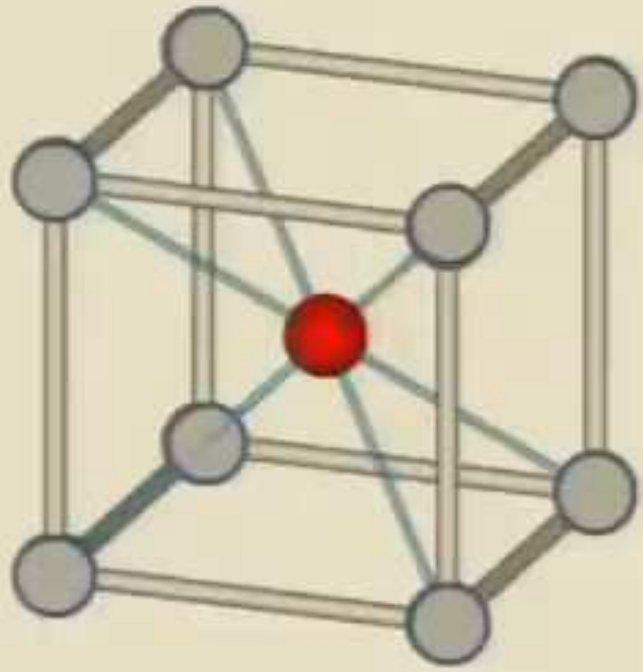
(More efficient sphere packing)





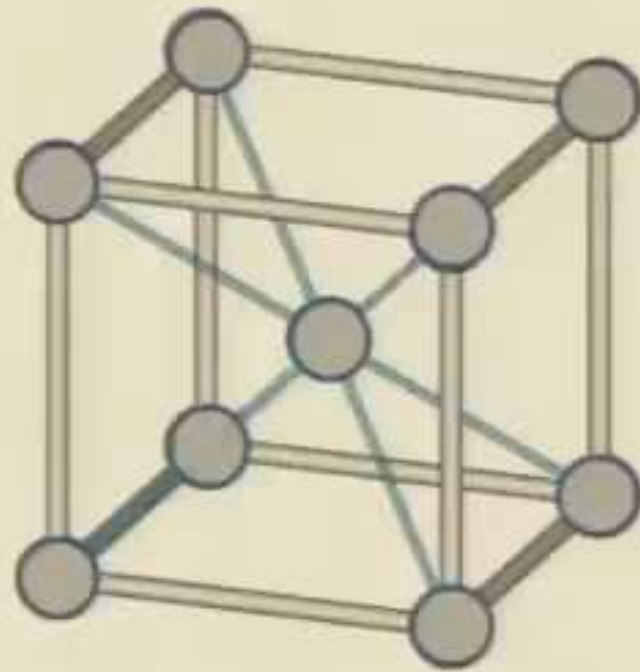
BCC Lattice





CsCl = Simple Cubic  
with Basis

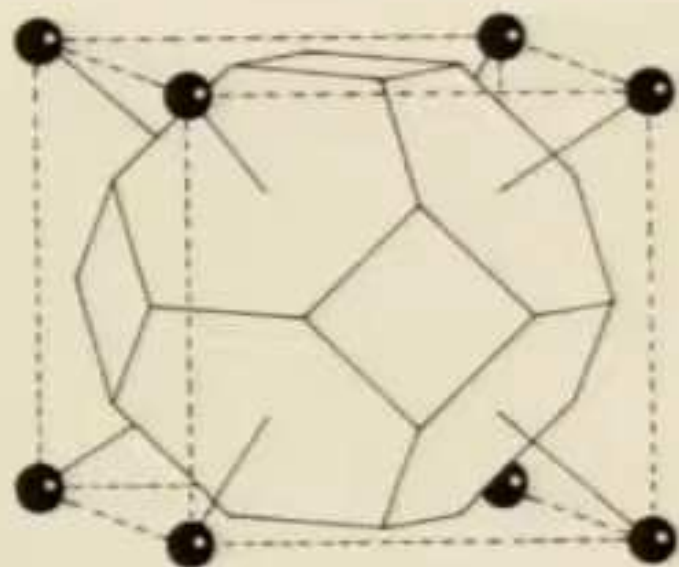
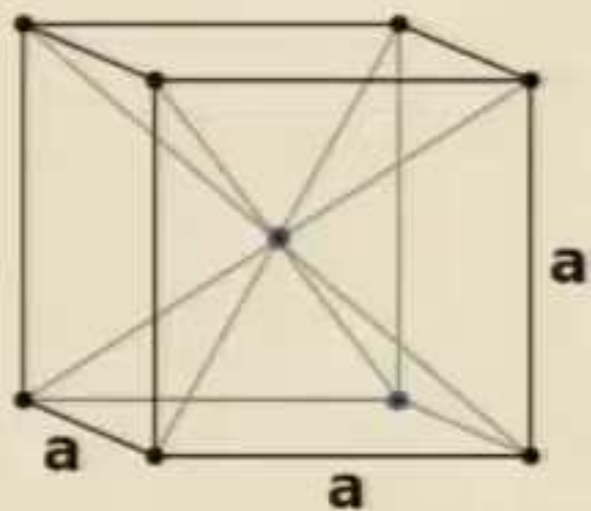
Cs at  $[0, 0, 0]$   
Cl at  $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$



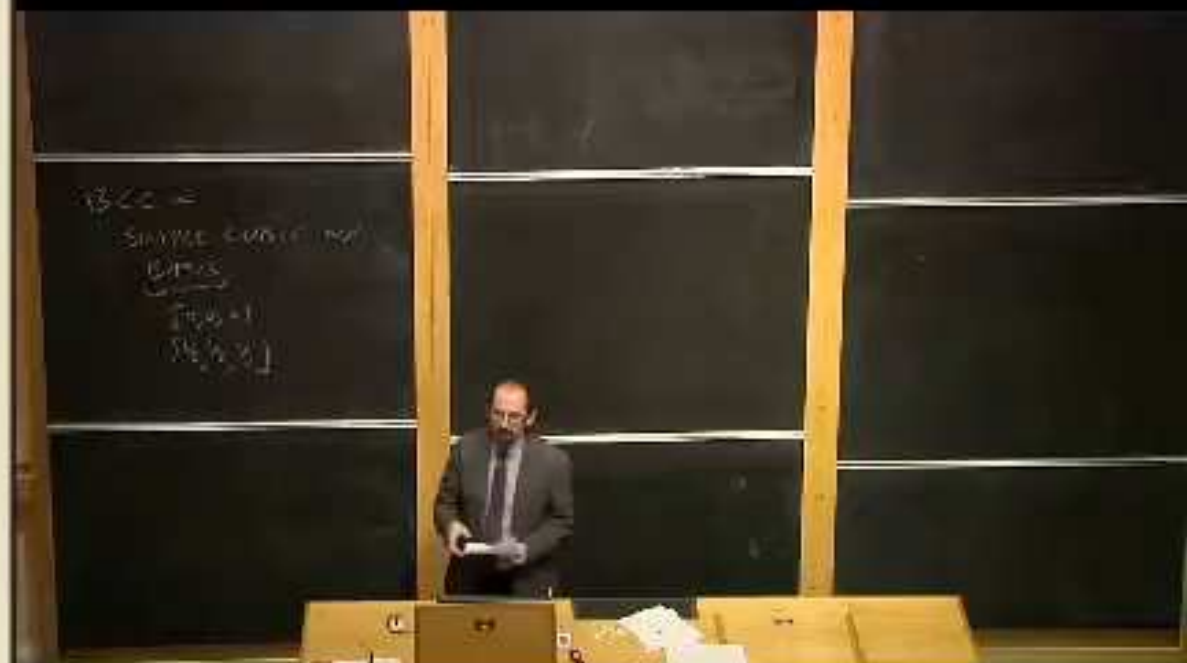
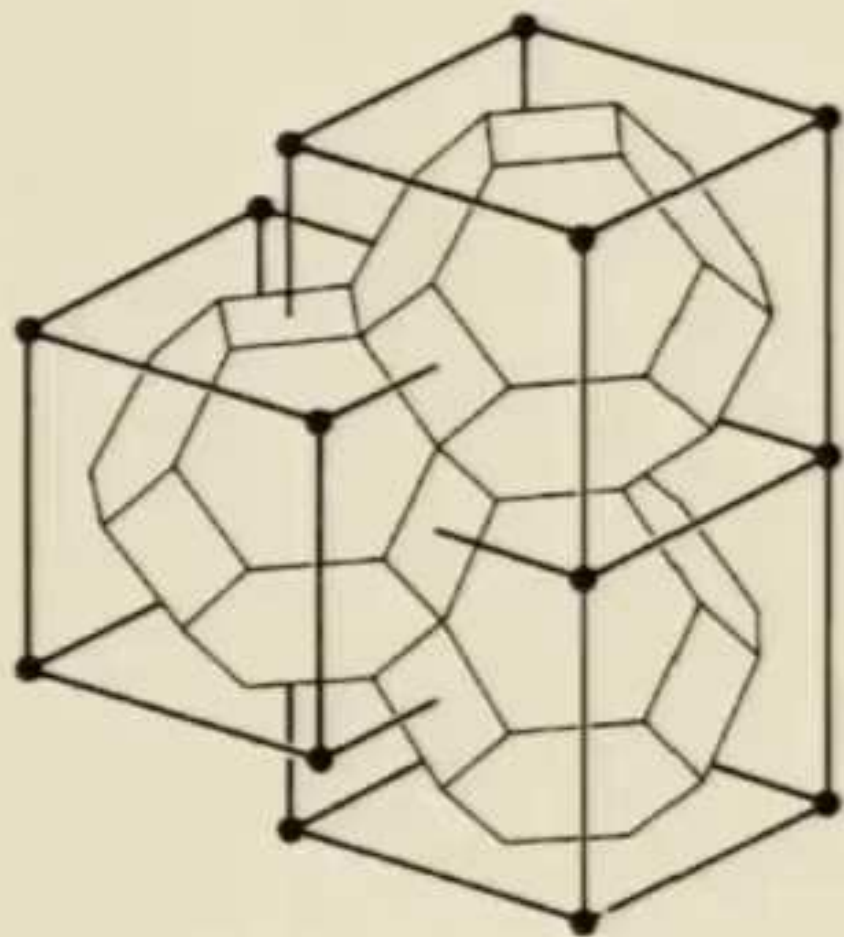
Cs = Simple Cubic  
with Basis

Cs at  $[0, 0, 0]$   
Cs at  $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$

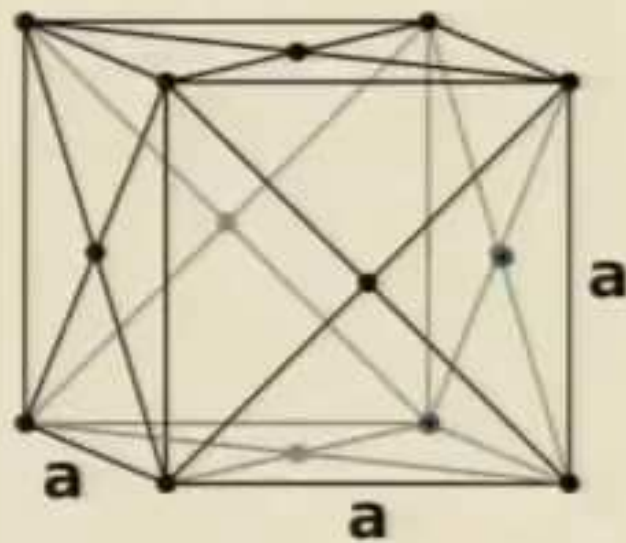




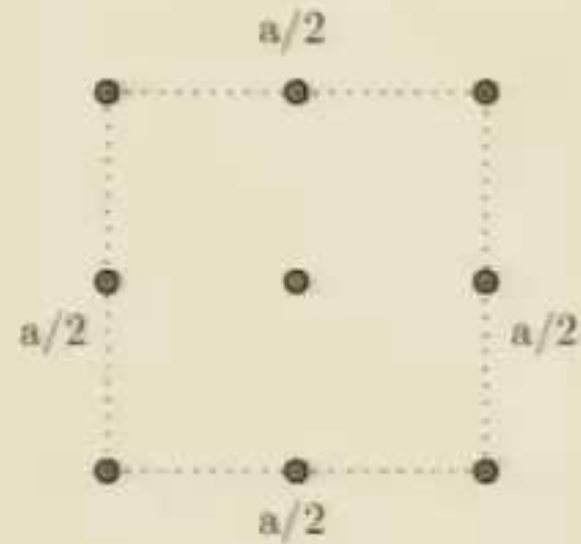
The Wigner-Seitz (Primitive)  
Unit Cell for the BCC lattice



# Unit cell of Face Centered Cubic Lattice (FCC) (Notated cubic-F)

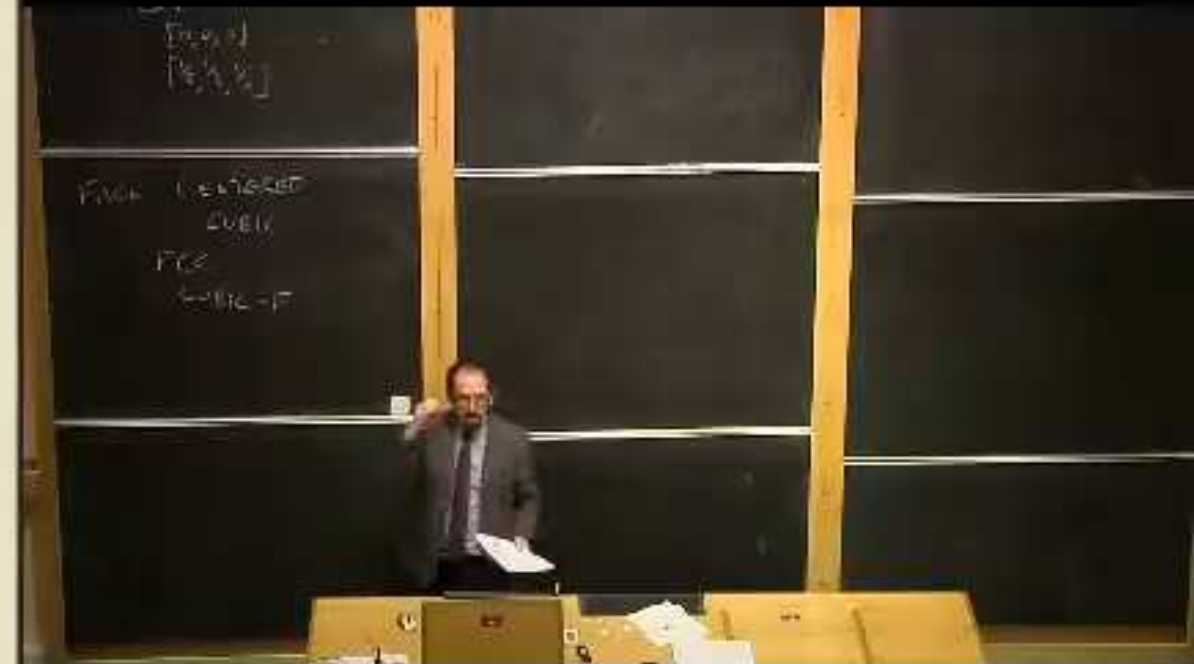


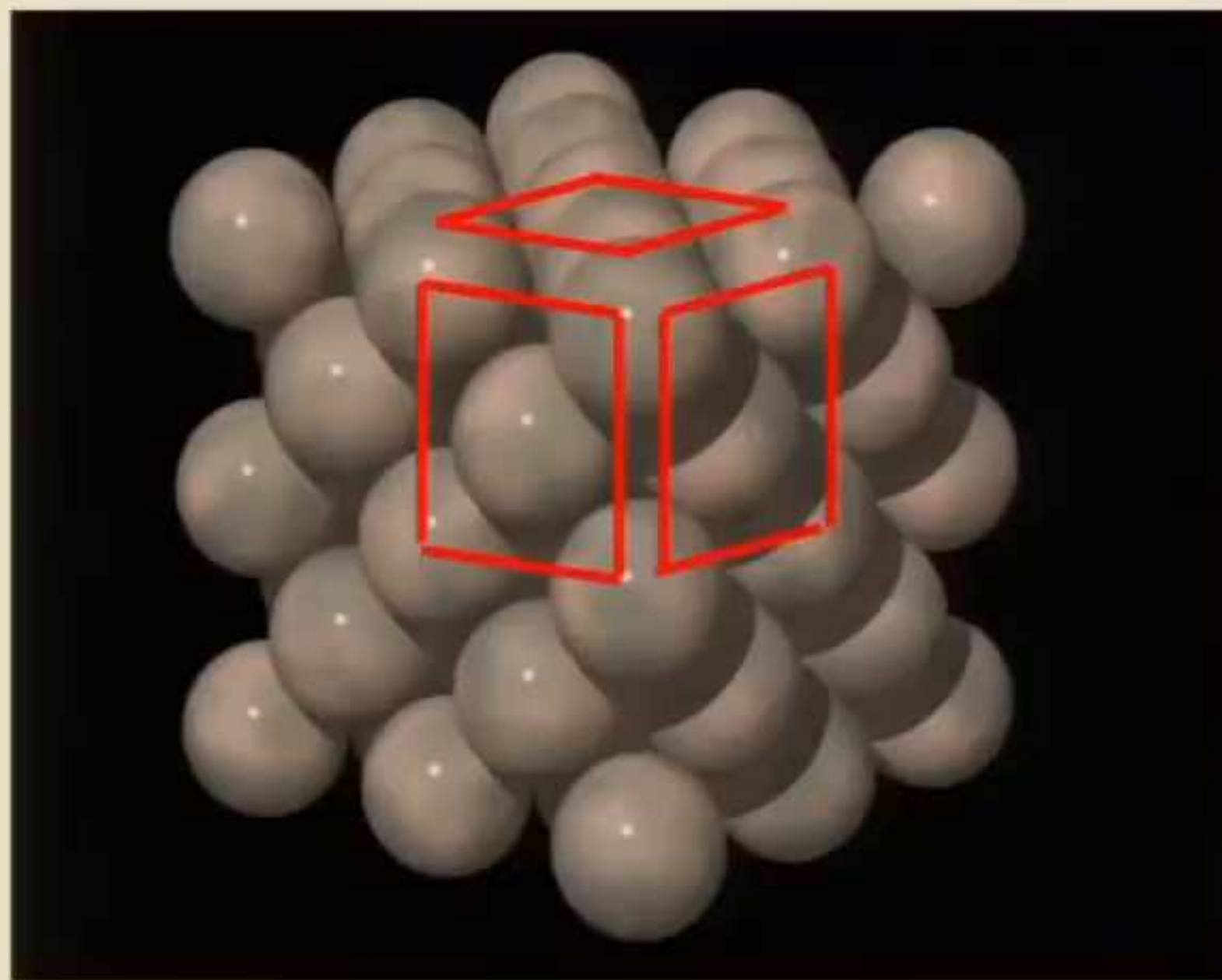
Conventional Unit Cell



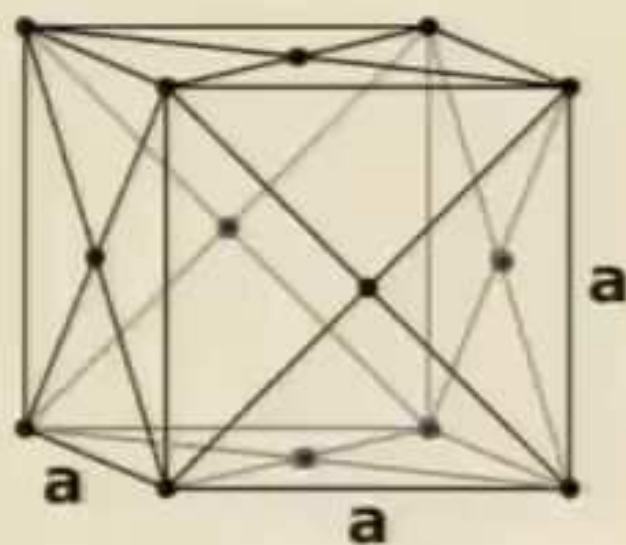
Plan View

(unlabeled points at height 0 and a)

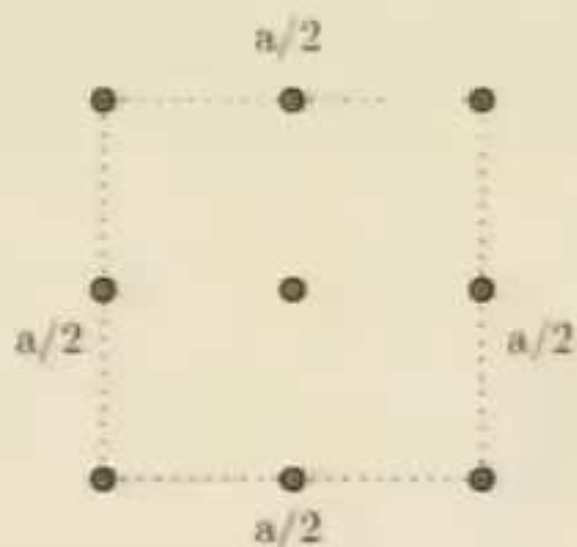




Unit cell of Face Centered Cubic Lattice (FCC) (Notated cubic-F)



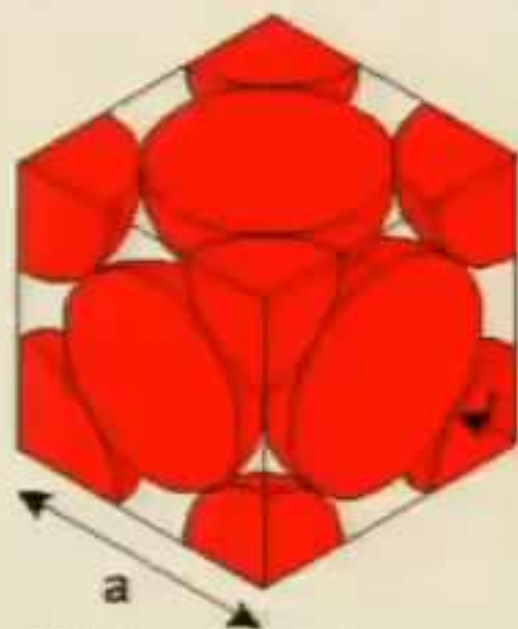
### Conventional Unit Cell



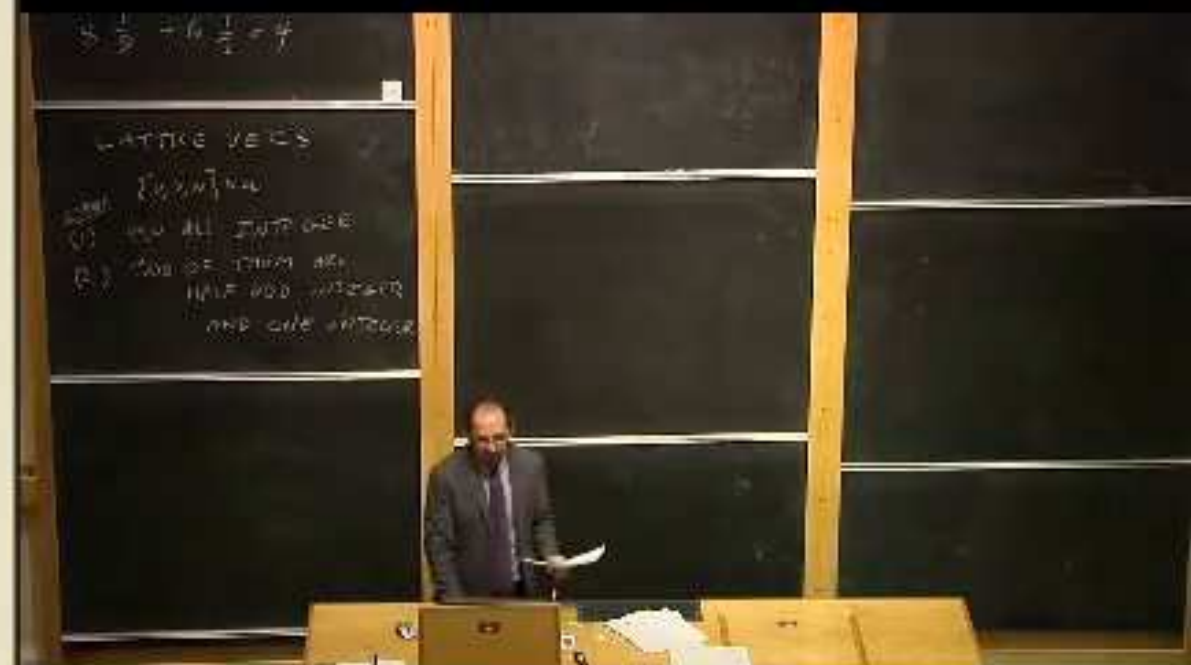
### Plan View

(unlabeled points at height 0 and  $a$ )

## FCC lattice

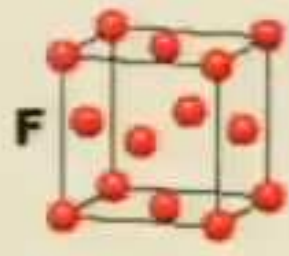
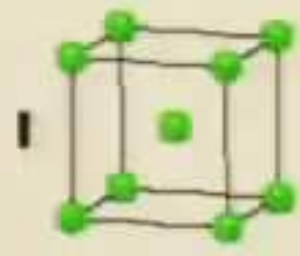
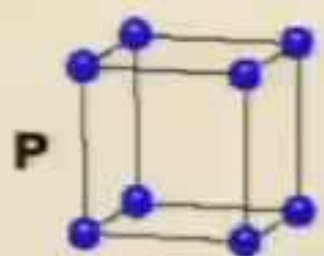


Most efficient sphere packing  
same density as "hcp" packing which  
we don't learn this year



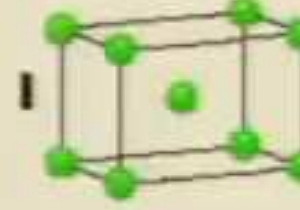
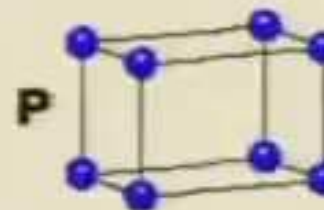
## CUBIC

$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$



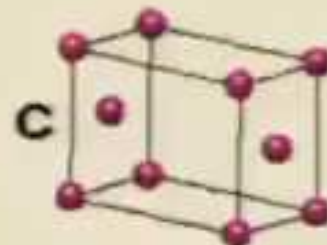
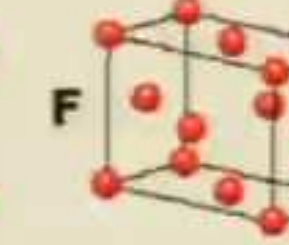
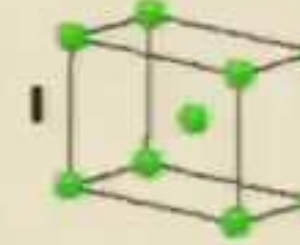
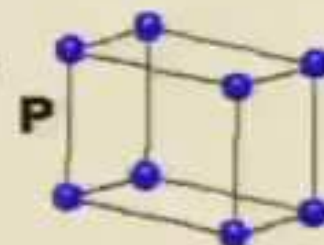
## TETRAGONAL

$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



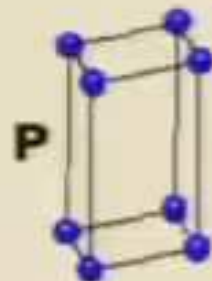
## ORTHORHOMBIC

$$a \neq b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



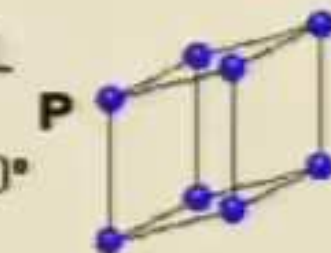
## HEXAGONAL

$$a = b \neq c$$
$$\alpha = \beta = 90^\circ$$
$$\gamma = 120^\circ$$



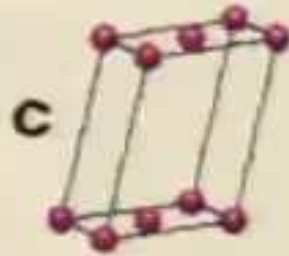
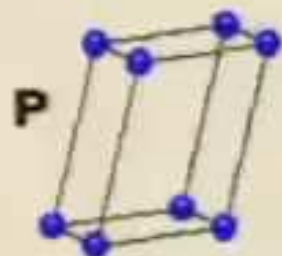
## TRIGONAL

$$a = b = c$$
$$\alpha = \beta = \gamma \neq 90^\circ$$



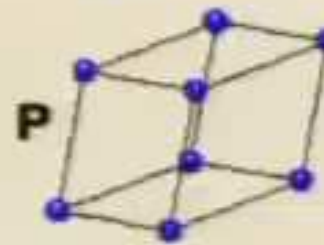
## MONOCLINIC

$$a \neq b \neq c$$
$$\alpha = \gamma = 90^\circ$$
$$\beta \neq 120^\circ$$



## TRICLINIC

$$a \neq b \neq c$$
$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



# The 14 Bravais Lattice Types

### 4 Types of Unit Cell

P = Primitive

I = Body-Centred

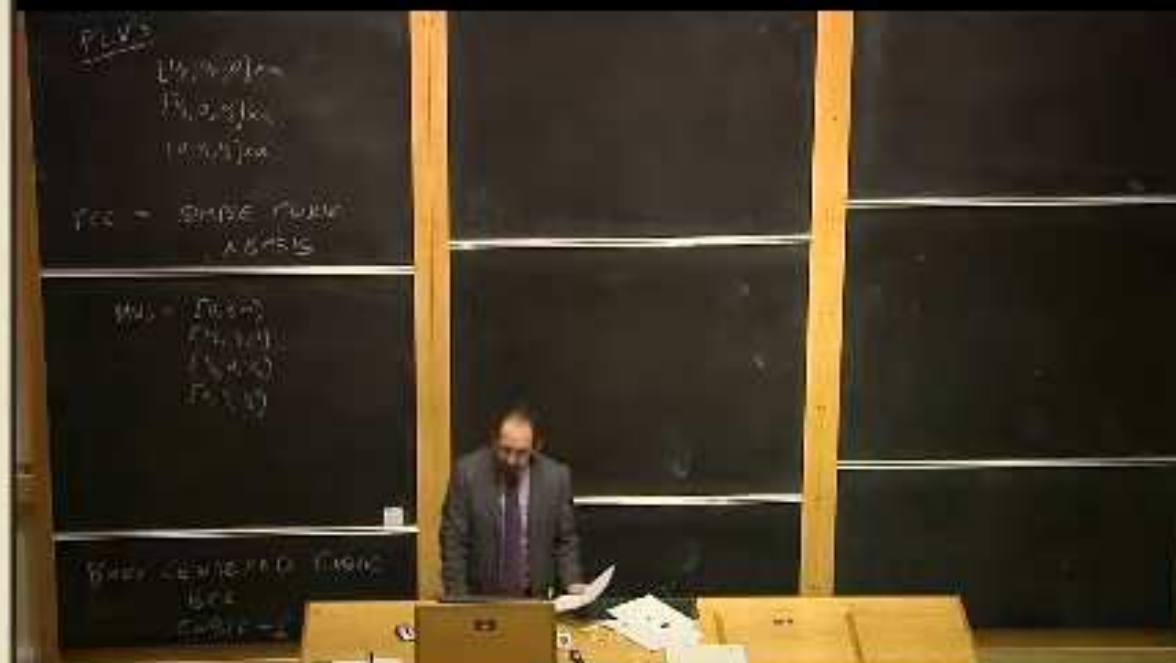
F = Face-Centred

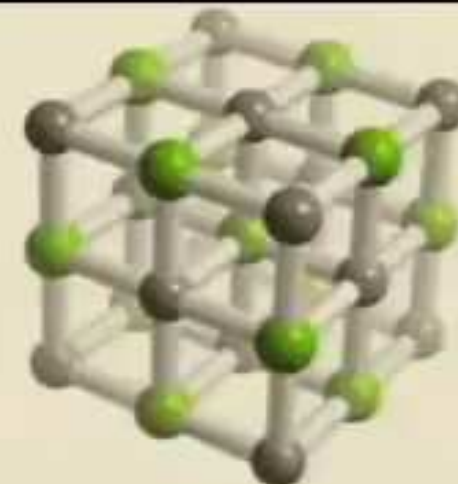
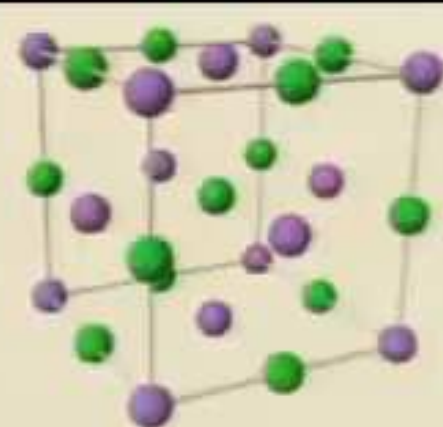
C = Side-Centred

+

7 Crystal Classes

→ 14 Bravais Lattices





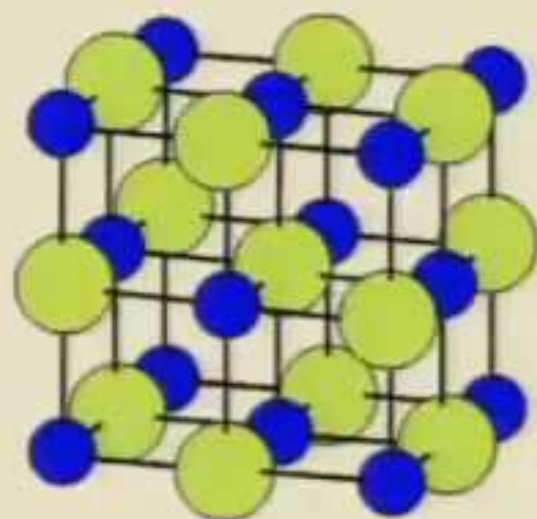
## sodium chloride (NaCl)

lattice: cubic F

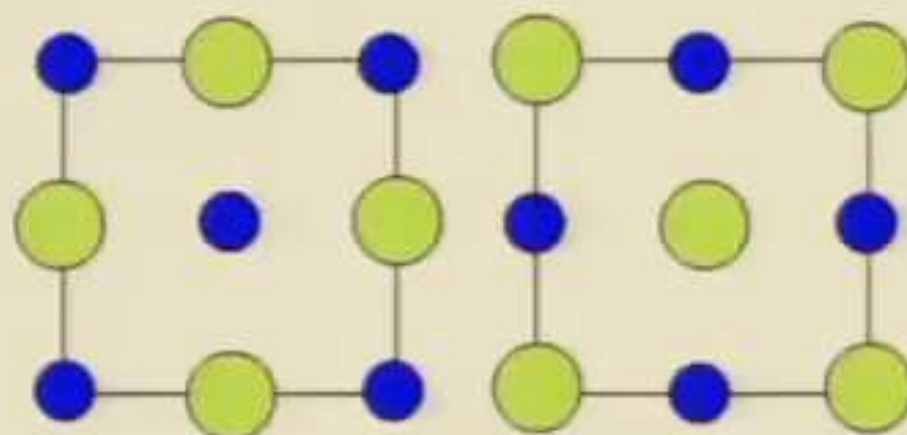
basis :

Na 000

Cl  $\frac{1}{2}\frac{1}{2}\frac{1}{2}$



Plan view

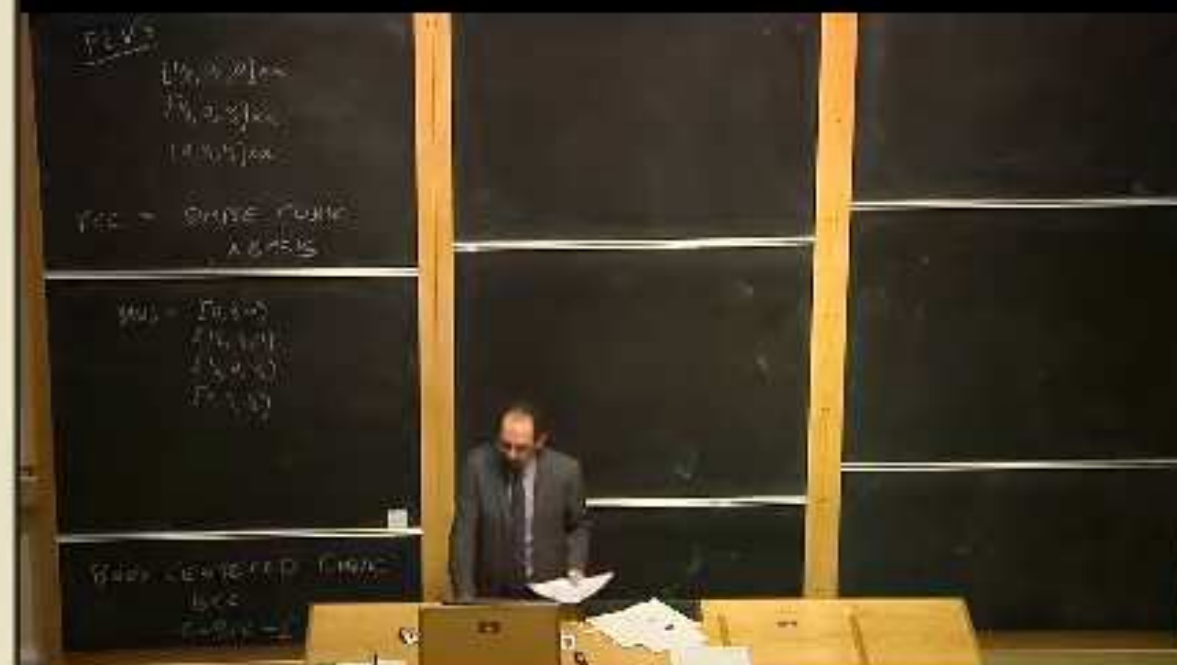


$z = 0$  layer

$z = \frac{1}{2}$  layer

Na forms FCC lattice.

Cl is displaced  $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$  from each Na



diamond (C) — also Si, Ge

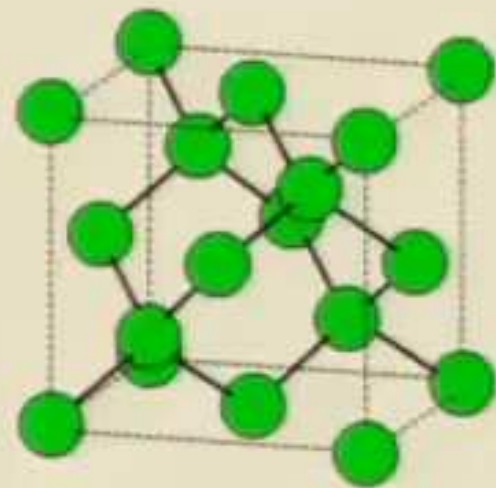
lattice: cubic F

basis :

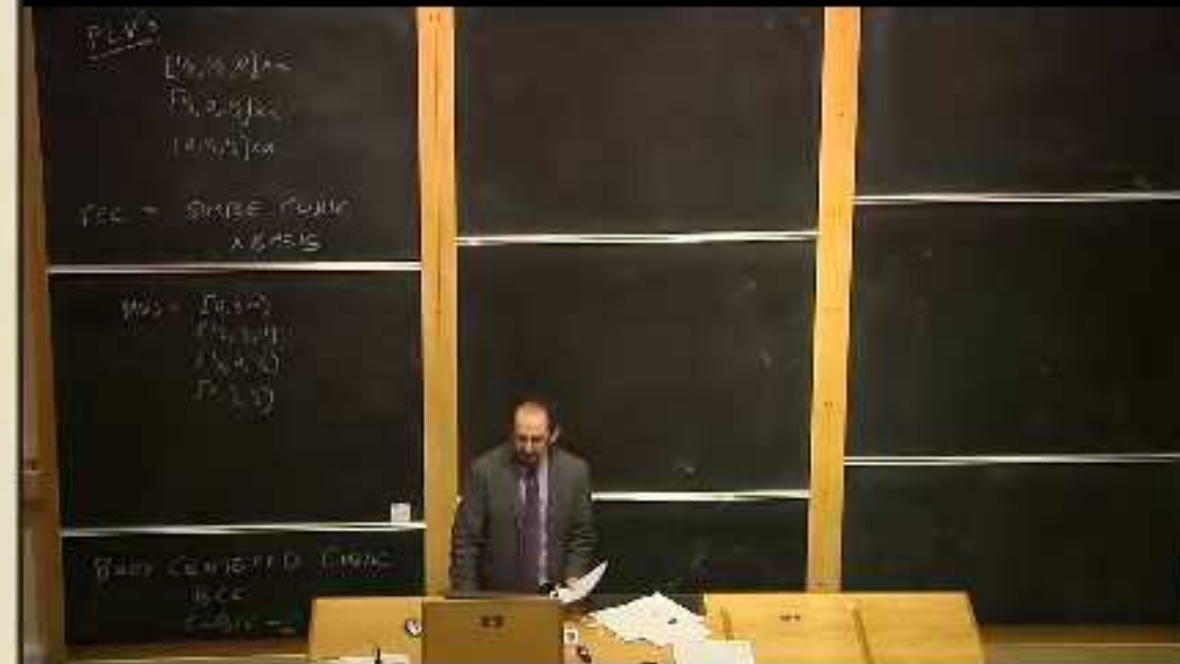
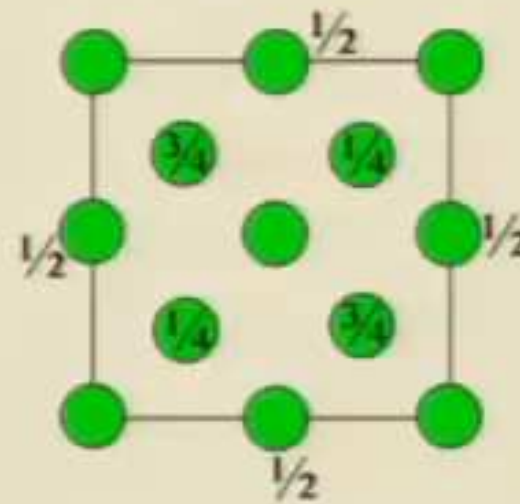
C 000



C  $\frac{1}{4}\frac{1}{4}\frac{1}{4}$

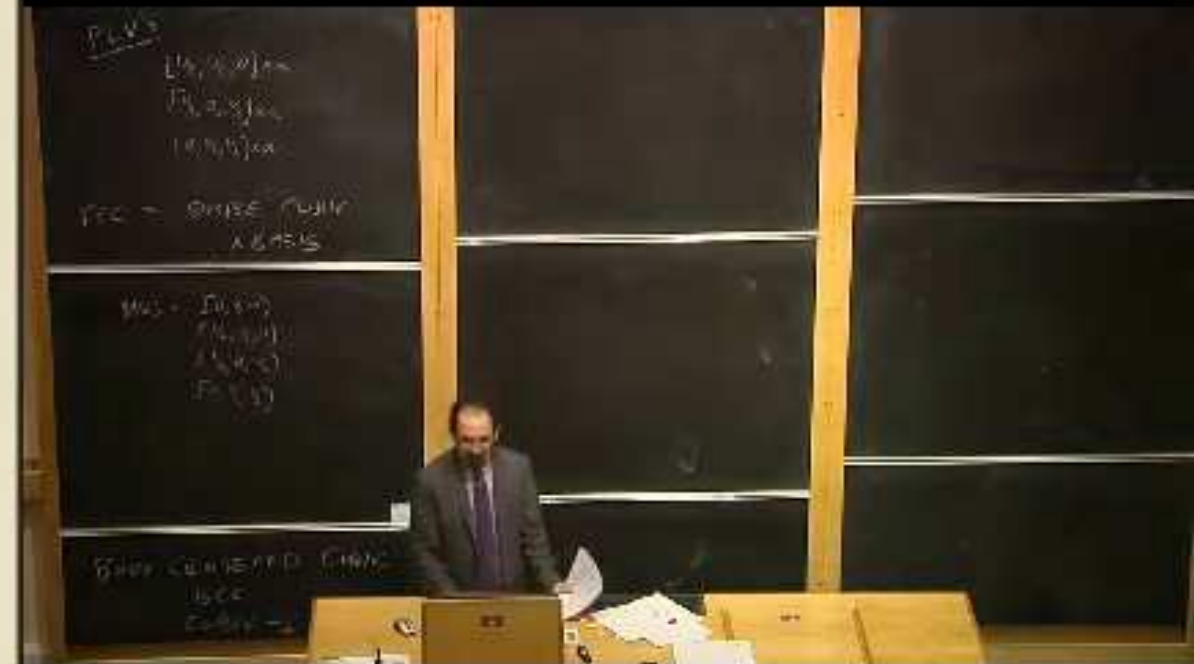
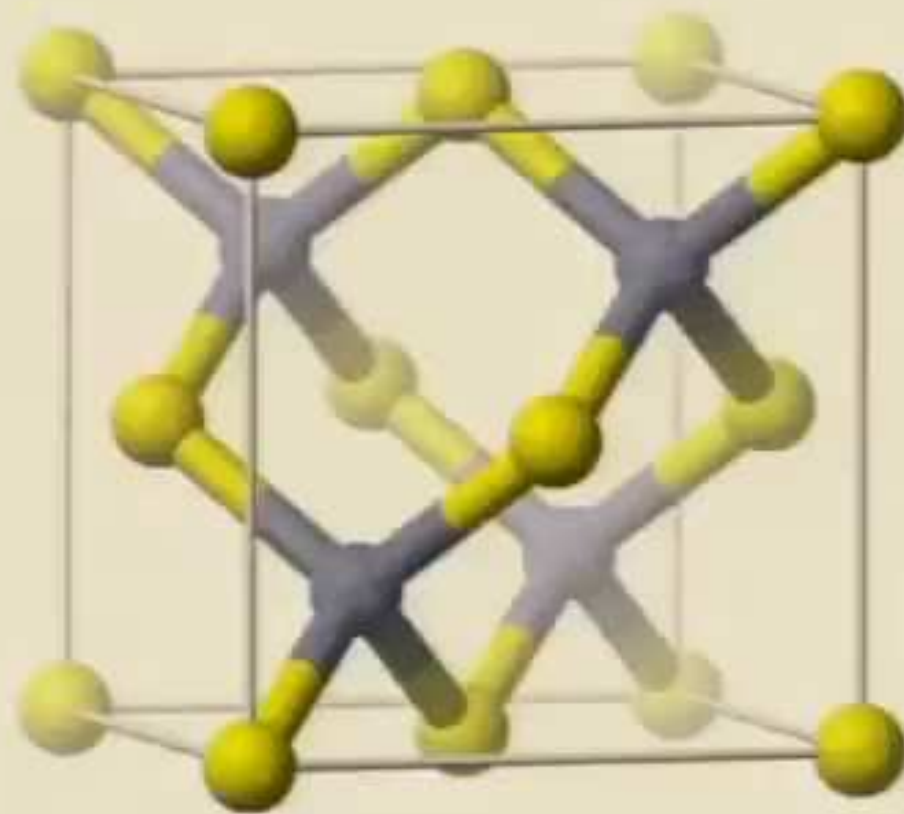


Plan view





## GaAs Structure...



PLVS

1/10/2014

1/10/2014

1/10/2014

VEC = 8/16 = 0.5  
ABTIS

MS = 1/16

1/16

1/16

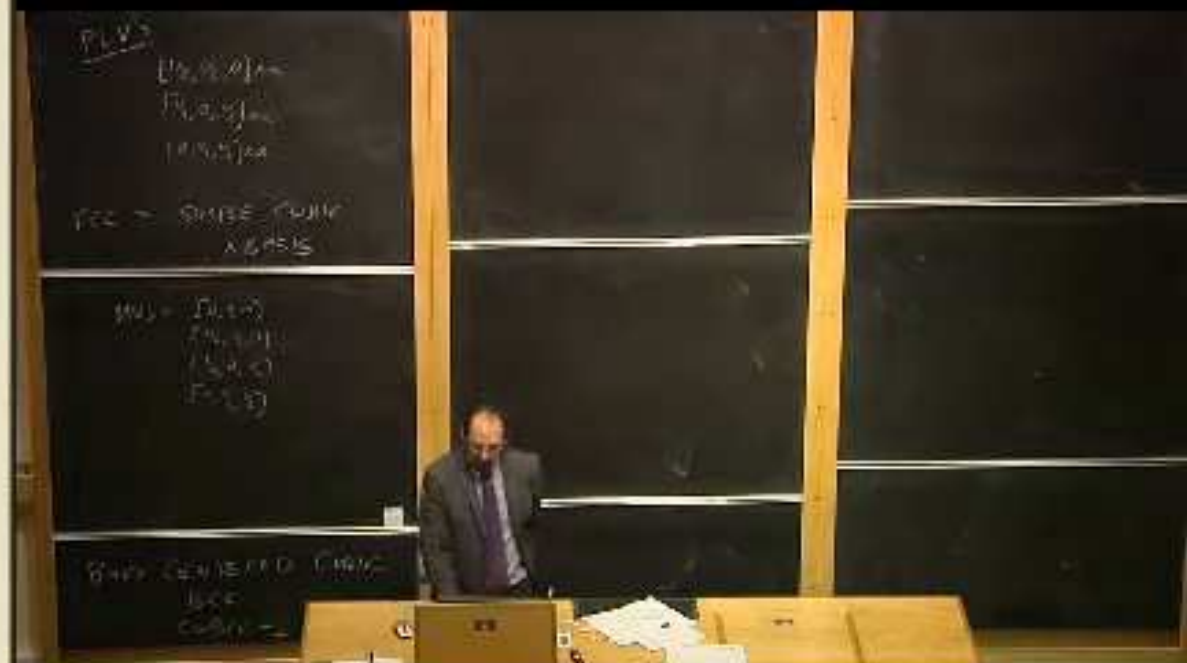
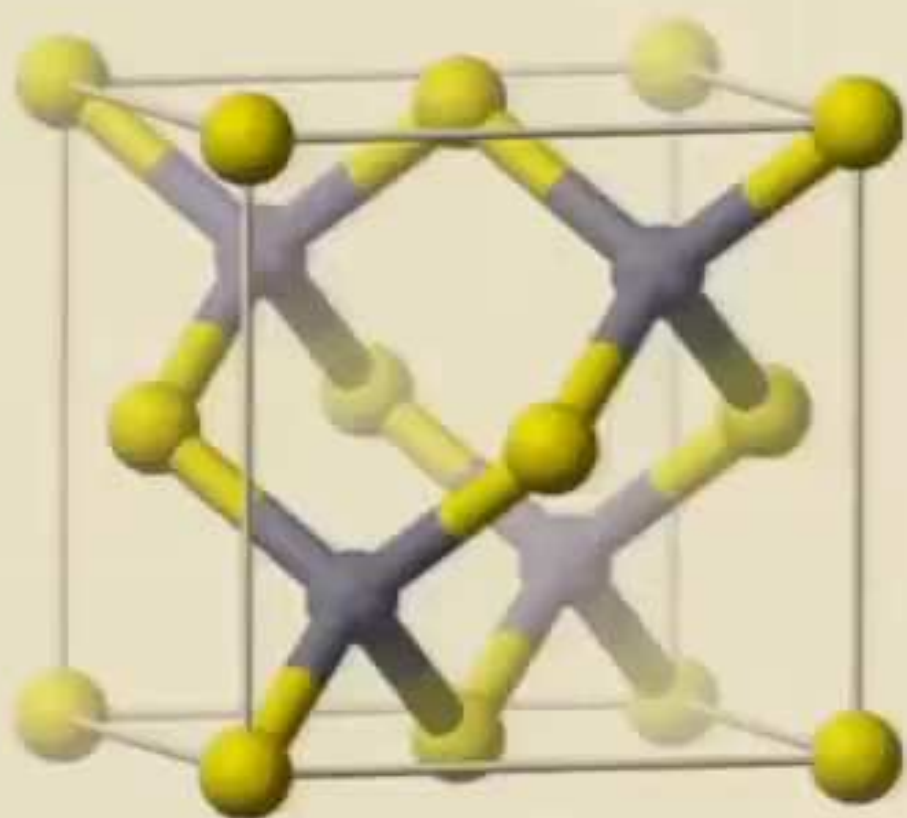
1/16

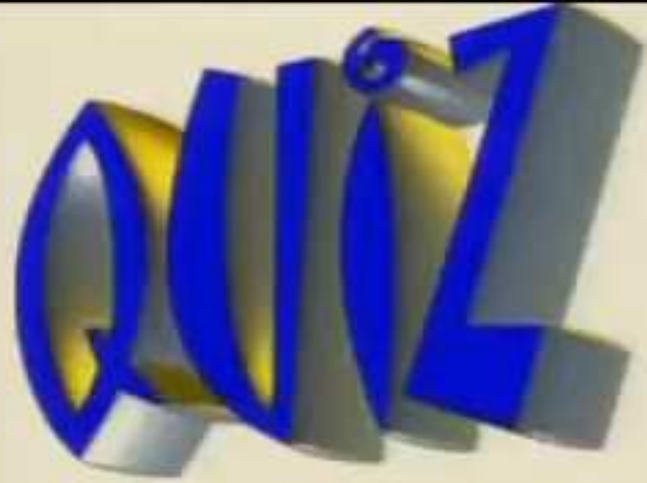
GROUP CENTERED CUBIC

1/16

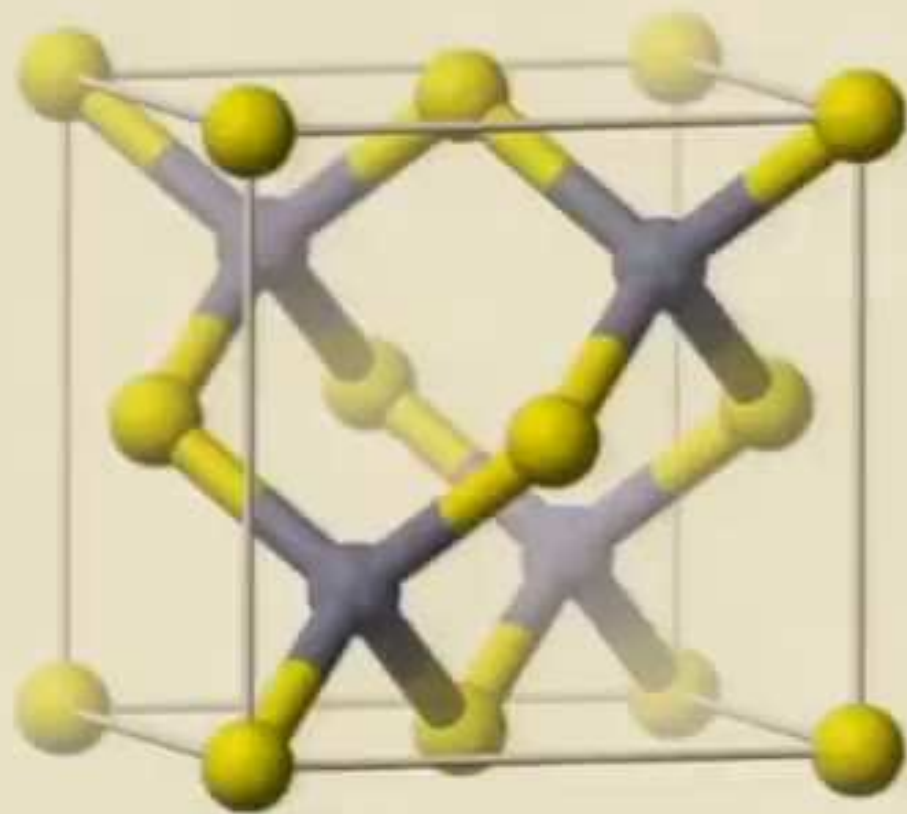
1/16

## GaAs Structure...





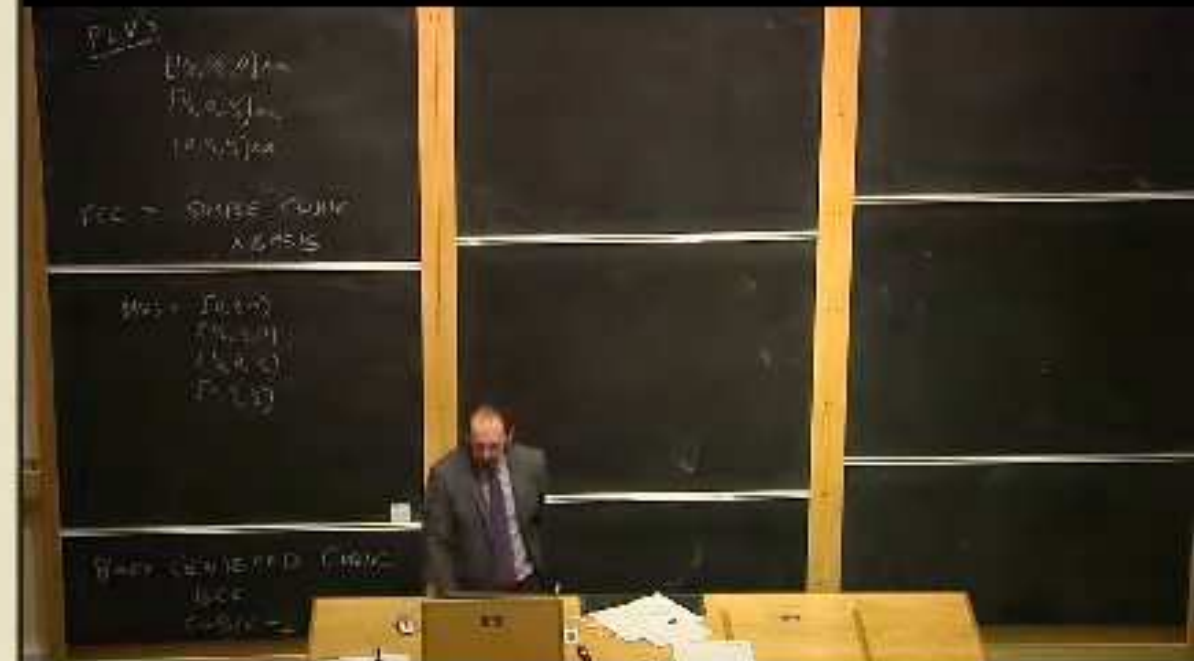
## GaAs Structure...



FCC:

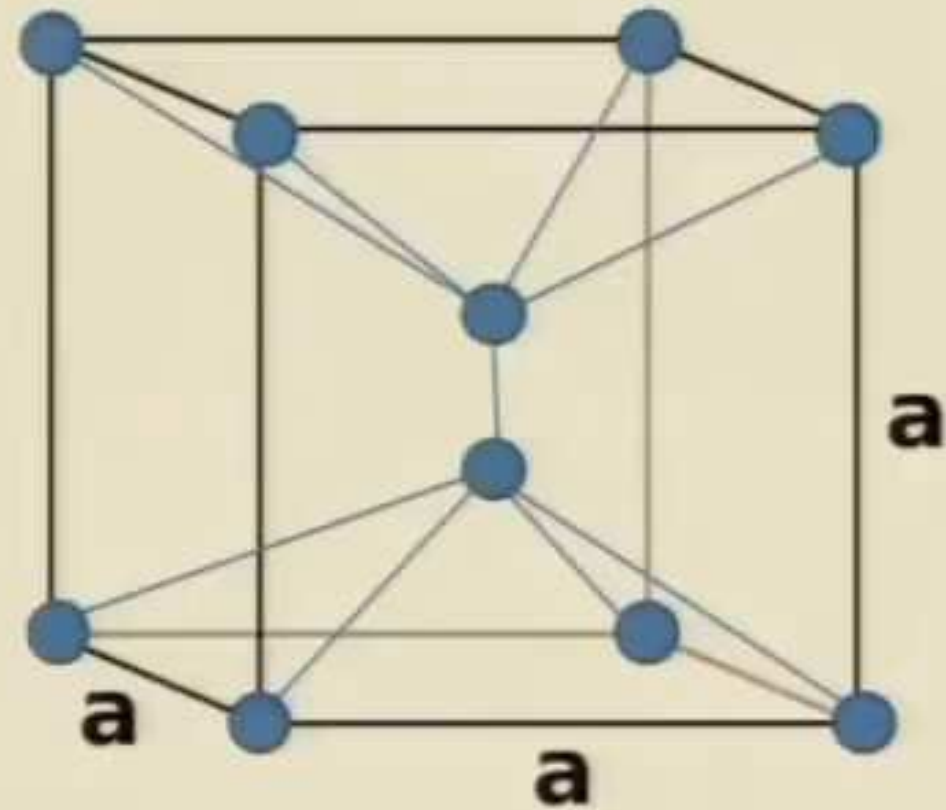
Basis: Yellow at  $(0,0,0)$

Blue at  $(1/4, 1/4, 1/4)$



Subtlety:

## Simple Cubic Unit Cell



This is not a simple cubic crystal?

(does not have the symmetry of simple cubic)

